Numerical Integration

Numerical integration is a process of evaluating a definite integral from set of tabulated values of the integrand $f(x)$. If the integrand is a function of single valued, then the numerical integration is known as Quadrature.

Simpson's one-third rule.

$$\int_{x_0}^{x_0+nh} f(x)dx = \frac{h}{3}\left[(y_0+y_n) + 4(y_1+y_3+\cdots+y_{n-1}) + 2(y_2+y_4+\cdots+y_{n-2})\right]$$

To apply Simpson's one-third rule, the given interval must be divided into even number of equal sub intervals.

Simpson's three-Eight rule:

$$\int_{x_0}^{x_0+nh} f(x)dx = \frac{3h}{8}\left[(y_0+y_n) + 3(y_1+y_2+y_4+y_5+\cdots+y_{n-4}+y_{n-2}+y_{n-1}) + 2(y_3+y_6+\cdots+y_{n-3})\right]$$

To apply this rule the given interval must be taken as multiple of 3.

Weddle's rule:

$$\int_{x_0}^{x_0+nh} f(x)dx = \frac{3h}{10}\left[(y_0+y_n) + 5(y_1+y_5+\cdots+y_{n-5}+y_{n-1}) + (y_2+y_4+\cdots+y_{n-4}+y_{n-2}) + 2(y_6+y_{12}+\cdots+y_{n-6}) + 6(y_3+y_9+\cdots+y_{n-3})\right]$$

While applying weddle's rule the number of subinterval should be taken as multiple of 6.

Weddle's rule for case $n=6$.

$$\int_{x_0}^{x_6} f(x)\,dx = \frac{3h}{10}\left[(y_0+y_6)+5(y_1+y_5)+(y_2+y_4)+6y_3\right]$$

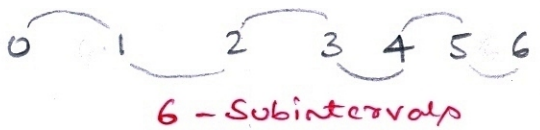
For $n=12$

$$\int_{x_0}^{x_{12}} f(x)\,dx = \frac{3h}{10}\left[(y_0+y_{12})+5(y_1+y_5+y_7+y_{11}) + (y_2+y_4+y_8+y_{10})+2y_6+6(y_3+y_9)\right]$$

Remember above two results for weddle's rule.

(1) Evaluate $\int_0^6 \frac{dx}{1+x^2}$ by using (a) Simpson's $\frac{1}{3}$ rule (b) Simpson's $\frac{3}{8}$ rule (c) weddle's rule.

Soln: Divide the interval $(0,6)$ into 6 equals parts each of width $h=1$.



Let $f(x)=\frac{1}{1+x^2}$.

x	0	1	2	3	4	5	6
$f(x)$	1	0.5	0.2	0.1	0.0588	0.0385	0.027
	y_0	y_1	y_2	y_3	y_4	y_5	y_6

(a) Simpson's $\frac{1}{3}$ rule

$$\int_0^6 \frac{1}{1+x^2}\,dx = \frac{h}{3}\left[(y_0+y_6)+4(y_1+y_3+y_5)+2(y_2+y_4)\right]$$

$$= \frac{1}{3}\left[(1+0.027)+4(0.5+0.1+0.0385)+2(0.2+0.058)\right]$$

$$= \frac{1}{3}\left[1.027+2.554+0.5176\right]$$

$$= 1.3662$$

(b) Simpson's $\frac{3}{8}^{th}$ rule

$$\int_0^6 \frac{1}{1+x^2}\,dx = \frac{3h}{8}\left[(y_0+y_6)+3(y_1+y_2+y_4+y_5)+2y_3\right]$$

$$= \frac{3}{8}\left[(1+0.027)+3(0.5+0.2+0.0588+0.0385)+2(0.1)\right]$$

$$= \frac{3}{8}\left[1.027+2.3919+0.2\right]$$

$$= 1.3571$$

(c) Weddle's rule

$$\int_0^6 \frac{1}{1+x^2}\,dx = \frac{3h}{10}\left[(y_0+y_6)+5(y_1+y_5)+(y_2+y_4)+6y_3\right]$$

$$= \frac{3}{10}\left[(1+0.027)+5(0.5+0.0385)+(0.2+0.0588)+6(0.1)\right]$$

$$= 1.3735.$$

(2) Calculate $\int_4^{5.2} \log x\,dx$ by using

(a) Simpson's $\frac{1}{3}$ rule (b) Simpson's $\frac{3}{8}$ rule (c) Weddle's rule.

We divide the given interval into 6 equal parts each of width $h=0.2$.

Soln:-

x	4	4.2	4.4	4.6	4.8	5.0	5.2
$\log x$	1.3862	1.4350	1.4816	1.5260	1.5686	1.6094	1.6486
	y_0	y_1	y_2	y_3	y_4	y_5	y_6

(a) Simpson's $\frac{1}{3}$ rule

$$\int_4^{5.2} \log x\,dx = \frac{h}{3}\left[(y_0+y_6)+4(y_1+y_3+y_5)+2(y_2+y_4)\right]$$

$$= \frac{0.2}{3}\left[(1.3862+1.6486)+4(1.4350+1.5260+1.6094)+2(1.4816+1.5686)\right]$$

$= \frac{0.2}{3}[3.0348 + 18.2816 + 6.1004]$

$= 1.8277$

(b) Simpson's $\frac{3}{8}$ rule

$$\int_4^{5.2} \log x \, dx = \frac{3h}{8}[(y_0+y_6) + 3(y_1+y_2+y_4+y_5) + 2y_3]$$

$$= \frac{3}{8}[0.2][(1.3862+1.6486) + 3(1.4350+1.4816 + 1.5686+1.6094) + (1.5260)2]$$

$$= \frac{0.6}{8}[3.0348 + 18.2838 + 3.052]$$

$= 1.8277$

(c) weddle's rule

$$\int_4^{5.2} \log x \, dx = \frac{3h}{10}[(y_0+y_6) + 5(y_1+y_5) + (y_2+y_4) + 6y_3]$$

$$= \frac{3(0.2)}{10}[(1.3862+1.6486) + 5(1.4350+1.6094) + (1.4816+1.5686) + 6(1.5260)]$$

$= 1.8277$

(2) using Simpson's $\frac{1}{3}$ rule, evaluate $\int_0^3 \frac{dx}{1+x}$ by dividing the interval $(0,3)$ into six equal parts.

Soln: Let $y = \frac{1}{1+x}$, $h = 0.5$

x	0	0.5	1	1.5	2	2.5	3
y	1	0.6666	0.5	0.4	0.3333	0.2857	0.250
	y_0	y_1	y_2	y_3	y_4	y_5	y_6

By Simpson's $\frac{1}{3}$ rule

$$\int_0^3 \frac{1}{1+x}\,dx = \frac{h}{3}\left[(y_0+y_6)+4(y_1+y_3+y_5)+2(y_2+y_4)\right]$$

$$= \frac{0.5}{3}\left[(1+0.25)+4(0.6666+0.4+0.2857)+2(0.5+0.3333)\right]$$

$$= 0.1666\left[1.25+5.4092+1.6666\right]$$

$$= 1.3870$$

④ Using Simpson's $\frac{3}{8}$ rule obtain an approximate value of $\int_0^{0.3} (2x-x^2)^{1/2}\,dx$.

Soln:- Let $y = \sqrt{2x-x^2}$ and $h = 0.05$,

x	0	0.05	0.1	0.15	0.2	0.25	0.3
y	0	0.3122	0.4359	0.5268	0.6	0.6614	0.7141

By Simpson's $\frac{3}{8}$th rule

$$\int_0^{0.3} (2x-x^2)^{1/2}\,dx = \frac{3h}{8}\left[(y_0+y_6)+3(y_1+y_2+y_4+y_5)+2y_3\right]$$

$$= \frac{3}{8}(0.05)\left[0+0.7141+3(0.3122+0.4359+0.6+0.6614)+2(0.5268)\right]$$

$$= \frac{0.15}{8}\left[0.7141+6.0825+1.0536\right]$$

$$= 0.1461$$

(5) Evaluate $\int_0^1 e^{-x^2} dx$ by using Simpson's $\frac{3}{8}$th rule taking seven ordinates.

(Seven ordinates is six sub intervals)

Soln:- Let $y = e^{-x^2}$

x	0	$\frac{1}{6}$	$\frac{2}{6}$	$\frac{3}{6}$	$\frac{4}{6}$	$\frac{5}{6}$	1
y	1	0.9726	0.8948	0.7788	0.6412	0.4994	0.3679

By Simpson's $\frac{3}{8}$th rule, we have

$$\int_0^1 e^{-x^2} dx = \frac{3h}{8}\left[(y_0+y_6) + 3(y_1+y_2+y_4+y_5) + 2y_3\right]$$

$$= \frac{3}{8}\times\frac{1}{6}\left[1 + 0.3679 + 3(0.9726+0.8948+0.6412 + 0.4994) + 2(0.7788)\right]$$

$$= \frac{1}{16}\left[1.3679 + 9.024 + 1.5576\right]$$

$$= 0.7468.$$

Gauss - Quadrature approach

Gauss-Legendre two point formula or Gauss two point formula

$$\int_{-1}^{1} f(x)\,dx = f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right)$$

Gauss - Legendre three point formula or Gauss three point formula

$$\int_{-1}^{1} f(x)dx = \frac{1}{9}\left[5f\left(-\sqrt{\frac{3}{5}}\right) + 8f(0) + 5f\left(\sqrt{\frac{3}{5}}\right)\right]$$

The above results are true and useful when the limit of integration are $[-1, 1]$.

The limits of integration for Gauss - Legendre integration rules are $[-1, 1]$. Therefore, we transform the limits $[a, b]$ to $[-1, 1]$ using a linear transformation.

$$\int_{-1}^{1} f(x)dx = \int_{a}^{b} f(x)$$

Let the transformation be $x = pt + q$

when $x = a$, we have $t = -1$, $a = -p + q$

when $x = b$, we have $t = 1$, $b = p + q$

Solving for p and q, we get

$$p = \frac{b-a}{2}, \quad q = \frac{b+a}{2}$$

$\therefore$ Required transformation is

$$x = \frac{1}{2}[(b-a)t + (b+a)]$$

$$dx = \frac{(b-a)}{2}\,dt$$

(1) Evaluate $\int_{-1}^{1} \frac{dx}{\sqrt{1+x^2}}$ by Gauss-Legendre 2 point formula.

Soln: Here $f(x) = \frac{1}{\sqrt{1+x^2}}$

Let $I = \int_{-1}^{1} f(x)\,dx$

Since the given limits of integration are in standard form, by Gauss-Legendre two point formula we have

$$I = f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right)$$

$$= 0.8660 + 0.8660$$

$$= 1.7320$$

$$f\left(-\frac{1}{\sqrt{3}}\right) = \frac{1}{\sqrt{1+\left(-\frac{1}{\sqrt{3}}\right)^2}}$$

$$= 0.8660$$

(2) Evaluate the integral $\int_{1}^{2} \frac{2x}{1+x^4}\,dx$ using Gauss two point and three point formula.

Soln: we reduce the interval $[1,2]$ to $[-1,1]$ to apply Gauss rules

lower limit $\leftarrow a = 1$

upper limit $\rightarrow b = 2$

By using the transformation

$$x = \frac{1}{2}[(b-a)t + (b+a)], \text{ we get}$$

$$x = \frac{1}{2}[(2-1)t + (2+1)]$$

$$x = \frac{t+3}{2}$$

$$dx = \frac{dt}{2}$$

$$\therefore \int_1^2 \frac{2x}{1+x^4}\,dx = \int_{-1}^{1} \frac{\frac{2(t+3)}{2}}{1+\left[\frac{(t+3)}{2}\right]^4}\,\frac{dt}{2}$$

$$= \int_{-1}^{1} \frac{8(t+3)}{[16+(t+3)^4]}\,dt$$

$$= \int_{-1}^{1} f(t)\,dt$$

Let $I = \int_{-1}^{1} f(t)\,dt$ where $f(t) = \frac{8(t+3)}{16+(t+3)^4}$

Using Gauss two point rule

$$I = f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right)$$

$$= f(-0.5773) + f(0.5773)$$

$$= 0.3841 + 0.1591$$

$$= 0.5433$$

$f(t) = \frac{8(t+3)}{16+(t+3)^4}$

Put $t=-0.5773$ we get $t = 0.5773$ given 0.1591

Using Gauss three point rule

$$I = \frac{1}{9}\left[5f\left(-\sqrt{\frac{3}{5}}\right) + 8f(0) + 5f\left(\sqrt{\frac{3}{5}}\right)\right]$$

$$= \frac{1}{9}\left[5f(-0.7745) + 8f(0) + 5f(0.7745)\right]$$

$$= \frac{1}{9}\left[5(0.4392) + 8(0.2474) + 5(0.1378)\right]$$

$$= 0.5404.$$

(3) Using Gauss three point formula Evaluate $\int_0^1 \frac{dx}{1+x}$.

Soln:- we reduce the interval $[0,1]$ to $[-1,1]$ to apply the Gauss-legendre three point rule.

we use the transformation

Here $a=0$ (lower limit), $b=1$ (Upper limit)

$$x = \frac{1}{2}[(b-a)t + (b+a)]$$

$$x = \frac{1}{2}[(1-0)t + (1+0)]$$

$$x = \frac{t+1}{2}$$

$$dx = \frac{dt}{2}$$

$$\therefore \int_0^1 \frac{dx}{1+x} = \int_{-1}^{1} \left(\frac{dt}{2}\right) \times \frac{1}{1+\left(\frac{t+1}{2}\right)}$$

$$= \int_{-1}^{1} \frac{1}{t+3}\, dt$$

$$= \int_{-1}^{1} f(t)\, dt$$

Let $I = \int_{-1}^{1} f(t)\, dt$, where $f(t) = \frac{1}{t+3}$

Using the Gauss three point rule, we obtain

$$I = \frac{1}{9}\left[5f\left(-\sqrt{\tfrac{3}{5}}\right) + 8f(0) + 5f\left(\sqrt{\tfrac{3}{5}}\right)\right]$$

$$= \frac{1}{9}[5f(-0.7745) + 8f(0) + 5f(0.7745)]$$

$$= \frac{1}{9}[5(0.4493) + 8(0.3333) + 5(0.2649)]$$

$$= 0.6930$$